Yoshitaka Sakamoto

Staff scientist, Cancer Genomics & Bioinformatics

Tokyo, Japan yos-sk

Summary

Computational biologist working on long-read sequencing and haplotype-resolved analysis of cancer genomes.

Education

The University of Tokyo, Ph.D. (Computational Biology), Department of Computational Biology and Medical Sciences, Graduate School of Frontier Sciences — 2020 – 2023

The University of Tokyo, MSc (Computational Biology), Department of Computational Biology and Medical Sciences, Graduate School of Frontier Sciences — 2018 – 2020

The University of Tokyo, BSc (Bioinformatics), Department of Bioinformatics and Systems Biology, Faculty of Science — 2014 – 2018

Experience

National Cancer Center Research Institute, Staff scientist — 2023 – present, 3 years

Japan Society for the Promotion of Science (JSPS), JSPS Research Fellow (DC2) — 2021 – 2023, 2 years
- Host institution: The University of Tokyo

Publications

Rare k-mers reveal centromere haplogroups underlying human diversity and cancer translocations — Oct 2026
Y. Shiraishi, Y. Ochi, M. Sugawa, **Y. Sakamoto**, et al.
doi.org/10.1016/j.cell.2026.08.053 (Cell)

Personalized reference genome-based pipeline reveals comprehensive haplotype-resolved views of cancer genomes — May 2026
First author
Y. Sakamoto, Y. Ochi, Y. Kogure, et al.
doi.org/10.64898/2026.05.28.728591 (bioRxiv)

mm2-ivh: simple and precise overlap detection in alpha satellite HORs with interval hashing — 2026
H. Suzuki, M. Sugawa, **Y. Sakamoto**, et al.
doi.org/10.1093/bioinformatics/btaf648 (Bioinformatics)

Targeted long-read methylation analysis using hybridization capture suitable for clinical specimens — 2025
K. Kunigo, S. Nagasawa, K. Kajiya, **Y. Sakamoto**, et al.
doi.org/10.1016/j.crmeth.2025.101215 (Cell Reports Methods)

Whole-genome sequencing reveals the molecular implications of the stepwise progression of lung adenocarcinoma 2023
Equal contribution (co-first author)
Y. Haga, **Y. Sakamoto**, K. Kajiya, et al.
doi.org/10.1038/s41467-023-43732-y (Nature Communications)

Long-Read Whole-Genome Sequencing Using a Nanopore Sequencer and Detection of Structural Variants in Cancer Genomes 2023
Y. Haga, **Y. Sakamoto**, M. Arai, et al.
doi.org/10.1007/978-1-0716-2996-3_13 (Nanopore Sequencing: Methods and Protocols (Methods in Molecular Biology))

Phasing analysis of lung cancer genomes using a long read sequencer 2022
First author
Y. Sakamoto, S. Miyake, M. Oka, et al.
doi.org/10.1038/s41467-022-31133-6 (Nature Communications)

Whole-genome Methylation Analysis of APOBEC Enzyme-converted DNA (~5 kb) by Nanopore Sequencing 2022
S. Zaha, **Y. Sakamoto**, S. Nagasawa, et al.
doi.org/10.21769/BioProtoc.4345 (Bio-protocol)

Long-read whole-genome methylation patterning using enzymatic base conversion and nanopore sequencing 2021
First author
Y. Sakamoto, S. Zaha, S. Nagasawa, et al.
doi.org/10.1093/nar/gkab397 (Nucleic Acids Research)

Application of long-read sequencing to the detection of structural variants in human cancer genomes 2021
First author
Y. Sakamoto, S. Zaha, Y. Suzuki, et al.
doi.org/10.1016/j.csbj.2021.07.030 (Computational and Structural Biotechnology Journal)

Long-read sequencing for non-small-cell lung cancer genomes 2020
First author
Y. Sakamoto, L. Xu, M. Seki, et al.
doi.org/10.1101/gr.261941.120 (Genome Research)

Single-cell sequencing techniques from individual to multiomics analyses 2020
Y. Kashima, **Y. Sakamoto**, K. Kaneko, et al.
doi.org/10.1038/s12276-020-00499-2 (Experimental & Molecular Medicine)

A new era of long-read sequencing for cancer genomics 2020
First author
Y. Sakamoto, S. Sereewattanawoot, A. Suzuki
doi.org/10.1038/s10038-019-0658-5 (Journal of Human Genetics)

MoMI-G: modular multi-scale integrated genome graph browser 2019
T. T. Yokoyama, **Y. Sakamoto**, M. Seki, et al.
doi.org/10.1186/s12859-019-3145-2 (BMC Bioinformatics)

Evaluation and application of RNA-Seq by MinION 2019
M. Seki, E. Katsumata, A. Suzuki, S. Sereewattanawoot, **Y. Sakamoto**, et al.
doi.org/10.1093/dnares/dsy038 (DNA Research)

Long read sequencing reveals a novel class of structural aberrations in cancers: identification and characterization of cancerous local amplifications 2019
First author
Y. Sakamoto, L. Xu, M. Seki, et al.
doi.org/10.1101/620047 (bioRxiv)

Identification of potential regulatory mutations using multi-omics analysis and haplotyping of lung adenocarcinoma cell lines 2018
S. Sereewattanawoot, A. Suzuki, M. Seki, **Y. Sakamoto**, et al.
doi.org/10.1038/s41598-018-23342-1 (Scientific Reports)

Oral Presentations

- **Telomere-to-telomere human genome construction and its application to cancer genome analysis** (invited educational lecture). The 85th Annual Meeting of the Japanese Cancer Association, Sep 2026.
- **Personalized reference genome-based cancer genome analysis applied to diverse cell lines**. The 85th Annual Meeting of the Japanese Cancer Association, Sep 2026.
- **Nearly complete personalized reference genome-based pipeline reveals comprehensive haplotype-resolved views of cancer genomes**. T2T-F2F, Sep 2026.
- **Cancer genome analysis based on personalized reference genome using long-read sequencing technologies** (invited). The 70th Annual Meeting of the Japan Society of Human Genetics, Dec 2025.
- **Reconstruction of cancer genome structure by personalized reference genome-based cancer genome analysis pipeline**. The 84th Annual Meeting of the Japanese Cancer Association, Sep 2025.
- **Personalized reference genome-based pipeline for cancer genome analyses to identify somatic mutations comprehensively**. 17th RECOMB Satellite Workshop on Computational Cancer Biology (RECOMB-CCB 2025), Apr 2025.
- **Development of a method for detecting DNA adducts using nanopore sequencing** (invited). 2024 Public Symposium of the Japanese Environmental Mutagen and Genome Society, Jun 2024.
- **Personalized matched control reference-based approach can detect structural variants accurately in cancer genomes**. The 82nd Annual Meeting of the Japanese Cancer Association, Sep 2023.

Skills

Bioinformatics
Long-read sequencing (ONT / PacBio), structural variant detection, haplotype phasing, de novo assembly, DNA methylation analysis

Programming
Python, Shell, Rust, R, workflow development (Snakemake)

Languages

Japanese
Native

English
Business